

MINERALOGY AND PETROLOGY OF APOLLO 12 SAMPLE No. 12013: A PROGRESS REPORT

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A mineralogical-petrological survey of Apollo 12 sample 12013 has been made with petrographic microscope and electron microprobe. The rock is extremely heterogeneous and consists of at least two separate fragmental units (light and dark) permeated by a once-fluid granitic component. The fragmental material includes a wide variety of lithic and crystal fragments some of which have not yet been reported from other lunar samples. The granitic component is essentially biminerale, with dominant potassic feldspar plus silica.

1. Introduction

Preliminary examination of Apollo 12 samples [1] has already revealed some interesting contrasts in chemistry, mineralogy, physical properties and ages with the Apollo 11 samples. As further analytical work proceeds, careful comparison of the two samples will form a basis for new sets of boundary conditions to be satisfied by proposed explanations of the origin of the lunar surface material. In the early stages of lunar surface exploration, individual samples which differ notably from the average in any way inherently assume a significance quite disproportional to their mass, e.g., the anorthositic component of the soil [2] and Luny Rock No. 1 [3] in the Apollo 11 sample. Apollo 12 sample no. 12013 [1] is chemically unique among all the lunar samples because of notable enrichments in SiO_2 (61 wt.%), K_2O (2.0 wt.%), ZrO_2 (0.30 wt.%), Li (100 ppm), Y (240 ppm), Ba (2150 ppm), Rb (33 ppm), Pb (30 ppm), Nb (170 ppm), Yb (20 ppm), Th (34 ppm), and U (11 ppm). Preliminary examination at the Lunar Receiving Laboratory revealed that the rock was extremely heterogeneous. The whole rock sample weighed 82 g and was approximately 4 cm in the longest dimension. Three polished thin sections were prepared from a thin slab cut several millimeters from the west end

(LRL orientation) of the specimen. Chips from a 4 mm thick slab (sample no. 12013, 10) immediately adjoining the thin section material were distributed for various chemical analyses via Dr. G.J. Wasserburg of the California Institute of Technology. Progress reports on some of these analyses are published in this volume. With such a heterogeneous sample, it is difficult, if not impossible, to interpret the chemical data without some knowledge of the mineralogy and petrography. The three thin sections currently being examined by us with petrographic microscope and electron microprobe represent a good cross-section of the material distributed for analysis, and, although our investigation is not complete, we herein report our preliminary results.

2. General description

The three thin sections are shown in figs. 1a–c. All locations refer to the grid on these figures. During mounting of the sections at LRL, the slab corresponding to section 12013, 15 was reversed relative to the other two. The three photographs show the present configuration of polished surfaces. The reversed numerical scale of fig. 1b is a qualitative reminder of the misorientation which must be kept in mind when

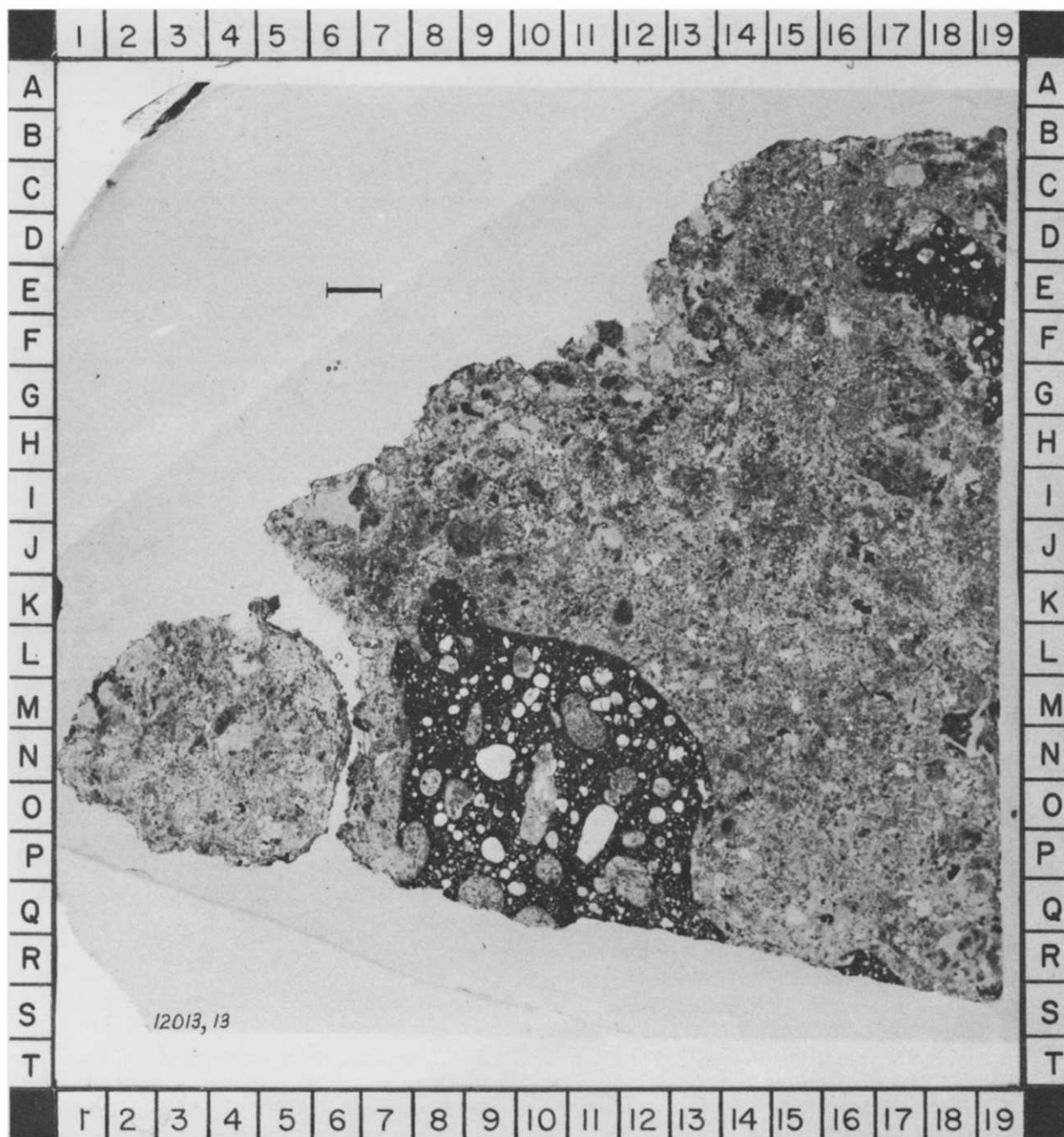


Fig. 1(a).

Fig. 1 (a)–(c). Transmitted light photographs of three thin sections forming a nearly continuous surface across sample 12013. The scale in each photograph is given by a 1 mm bar.

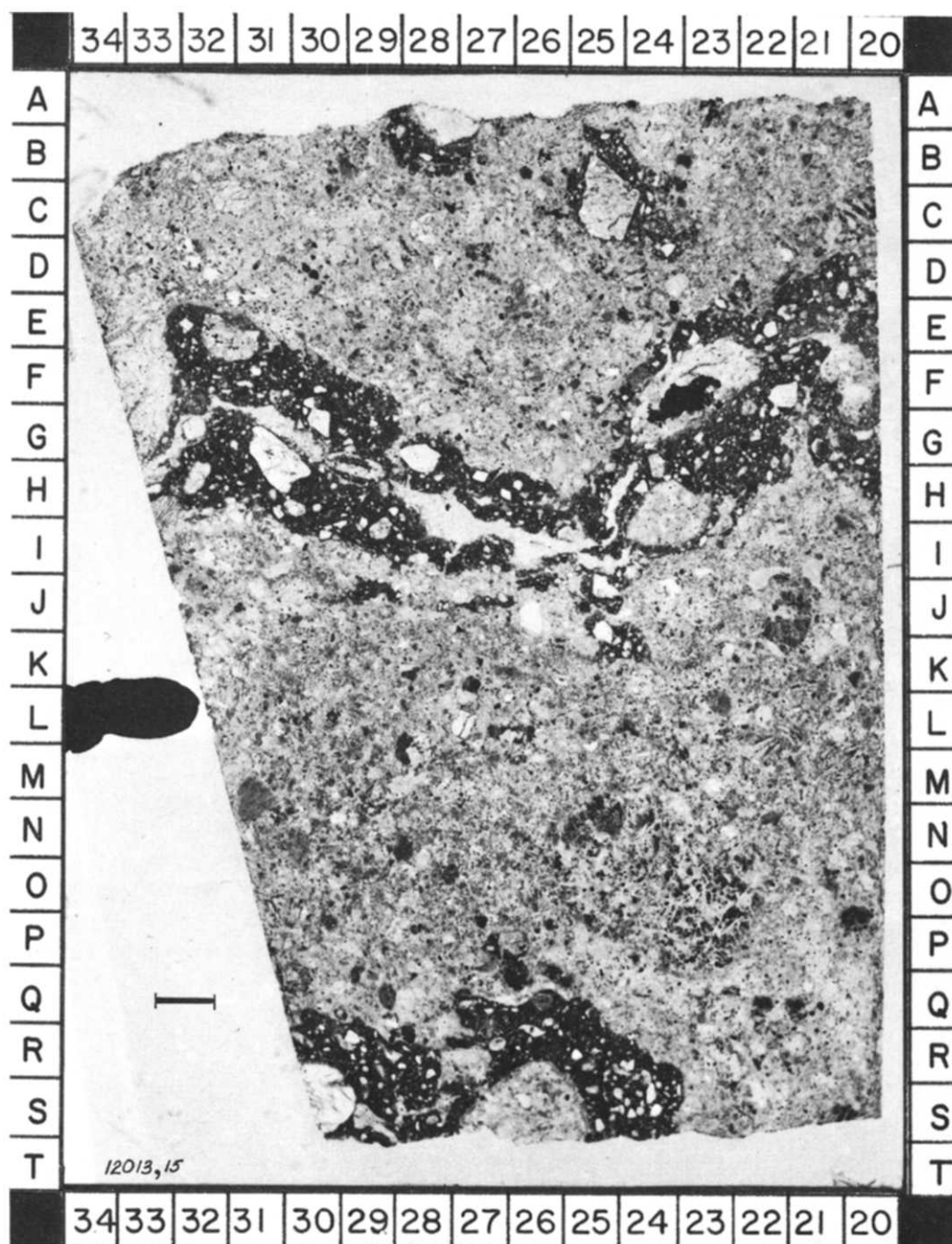


Fig. 1(b).

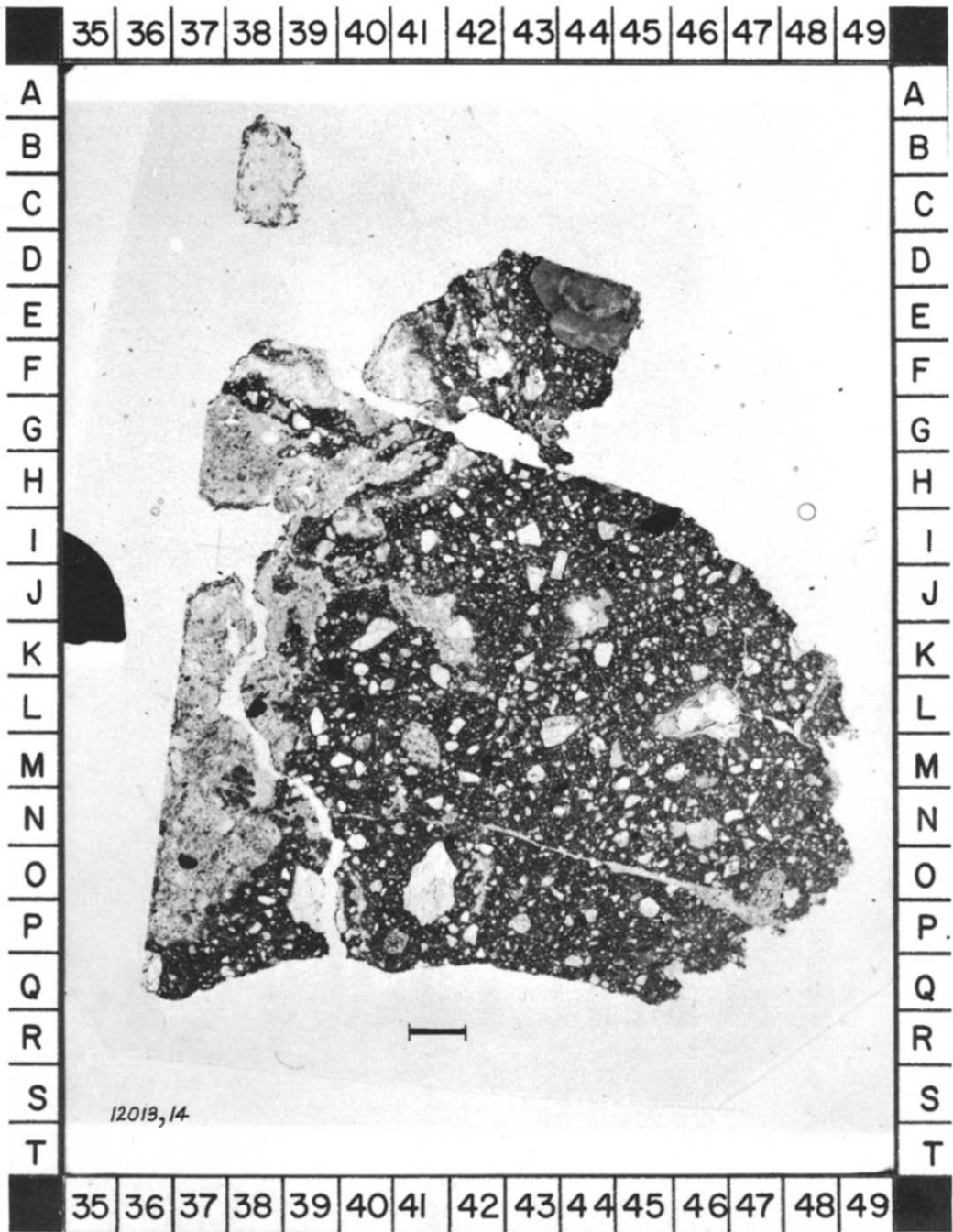


Fig. 1(c).

attempting to follow structures from one section to the next. Thus, corner 19/B of section 12013, 13 corresponds to corner 20/A of section 12013, 15 (neglecting kerf loss in sawing and thickness of original slab), and corner 30/S of 12013, 15 adjoins 36/Q of 12013, 14.

The most obvious large scale feature of 12013 is a clear-cut division between light and dark colored areas. The light and dark areas contain suites of fragmental material which appear to be distinct, and it is clear that at least two breccias are involved. The term polymict breccia was first proposed by Wahl [4] to describe meteorite material of similarly complex accumulation history. A somewhat less obvious feature is the presence of alkali feldspar and silica in various modes of occurrence throughout both light and dark areas. A vein rich in these two minerals may be seen cutting through the dark lithology from 47/P to 39/N in fig. 1c. Spherical to ovoid fillings of the same material are common in the dark area in fig. 1a. One of these (7/P) shows a connection to the main body of the light lithology. The light lithology throughout all three sections is permeated by and owes much of its light coloring to this alkali feldspar and silica-rich material. Its occurrence as cross-cutting veins and rounded fillings in the dark lithology as well as binding matrix in the light lithology (and to a lesser extent in the dark lithology) suggests that this silic material entered the rock in a fluid state, and that this "intrusive" event was the most recent in the formation of the rock.

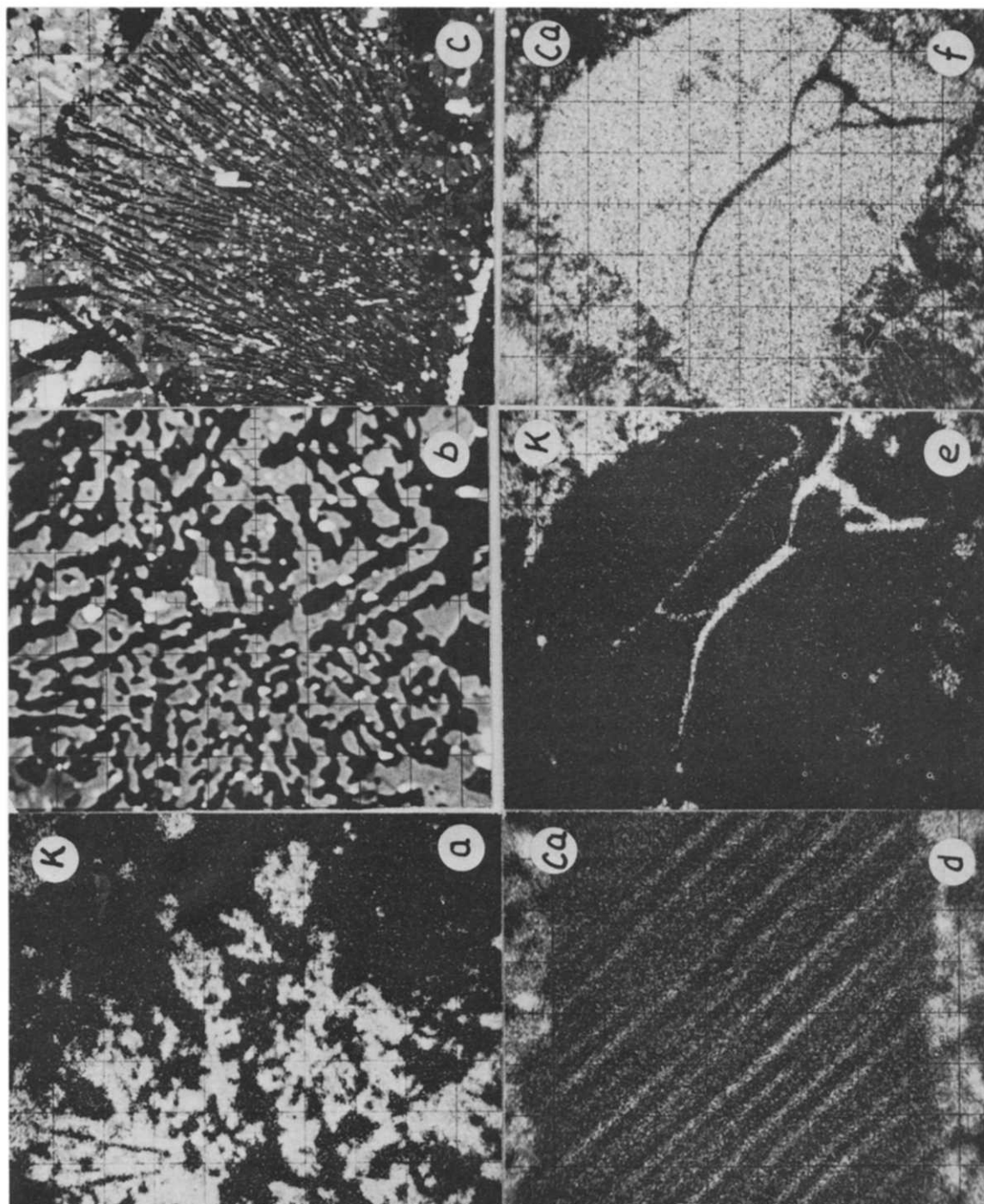
3. Light lithology

The light areas of 12013 can be described in terms of a fragmental component and a widespread potassic "granite" component which permeates much of the area and probably acts as the dominant cementing agent. A potassium X-ray scan across the light-dark matrix boundary at 10/L (fig. 2a) shows a well-defined drop in K concentration at the boundary. The granitic material did not permeate the dark lithology to the extent observed in the light areas, and we reason that the dark areas were lithified prior to the intrusion of the granitic phase. In contrast, the light lithology may have been relatively loose particulate material. A general description of the recognizable fragments is

given below. Quantitative mineralogical data are treated in a separate section.

Some of the particles in the light area are easily recognizable as fragments of rock types previously identified in other lunar samples. A fresh-looking fragment of sub-ophitic basalt is well displayed in area 23/O-P of fig. 1b. Other recognizable basaltic fragments, somewhat more altered, are found at 24-25/N-O, 30/Q, 20/M, 33/B, 18-19/M-N, 13/J-K, 11/K. Smaller fragments displaying basaltic texture and mineralogy are relatively common. The fresh-looking fragments contain little or no alkali feldspar plus silica, whereas altered fragments may contain as much as 5–10% of this granitic material.

A relatively coarse grained (up to 1 mm) igneous association of pyroxene, olivine, plagioclase, ilmenite and titanian chromite borders section 12013, 13 (5-13/C-N of fig. 1a). The coarse ferromagnesian silicates and subordinate plagioclase and opaques may represent an ultramafic igneous rock. A similar occurrence is located at 16-18/G-I. Some coarse-grained fragments of pyroxene and plagioclase mineralogy are found at 21/J, 32/K, 28/L-M, 27/D and 26/L. Certain shock features are found in this material as well as in some other lithic and crystal fragments throughout the rock. Microfractures and variable extinction are quite common. Fig. 3a illustrates lamellae in a large pyroxene grain which have been slightly deformed into an S-shaped array. Several plagioclase grains display linear trains of bubble-like inclusions (fig. 3b). Microprobe scans show no appreciable compositional contrast of these inclusions with the host. Incipient melting of plagioclase due to rapid heating by shock or some other process may have given rise to these features. Planar features in some pyroxene fragments are exsolution lamellae rather than shock features. One such example (taken from the dark lithology where they are also found) is illustrated by a Ca X-ray scan in fig. 2d showing Ca-rich lamellae in a Ca-poor host. In contrast to Apollo 11 breccias there is very little glass remaining in 12013, and we have found no clear-cut examples of diaplectic glass. This may be due to late-stage thermal annealing of the sample since many textures can be interpreted as the result of nearly complete devitrification. The rock sections are thicker than normal, making it difficult to identify small volumes of glass if they are present.



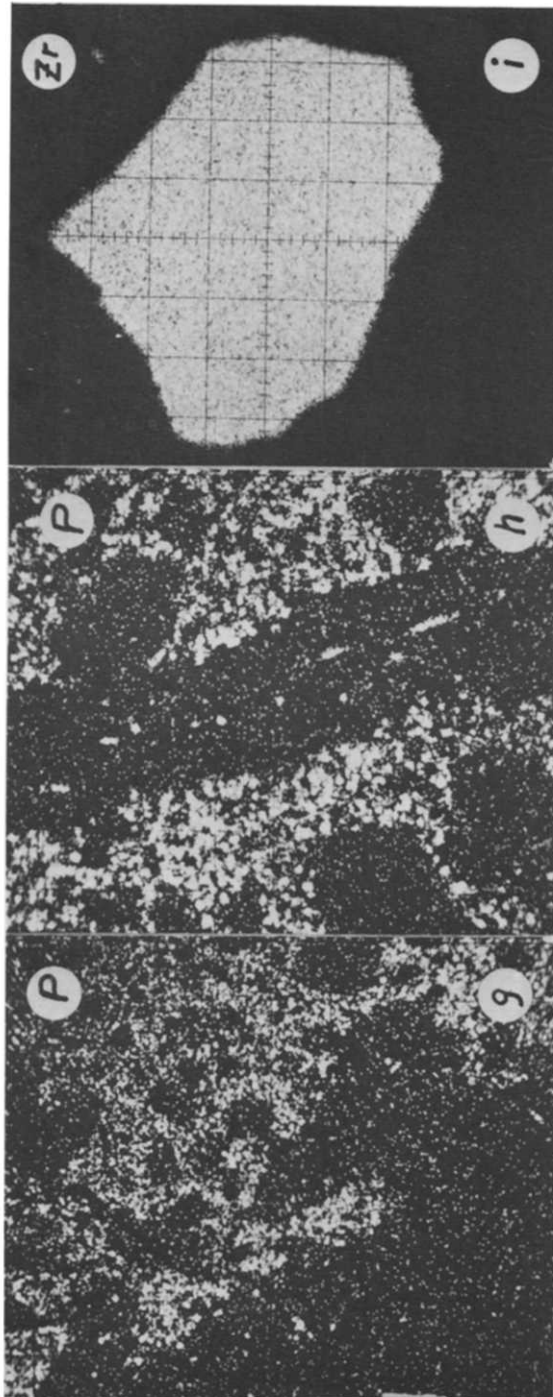
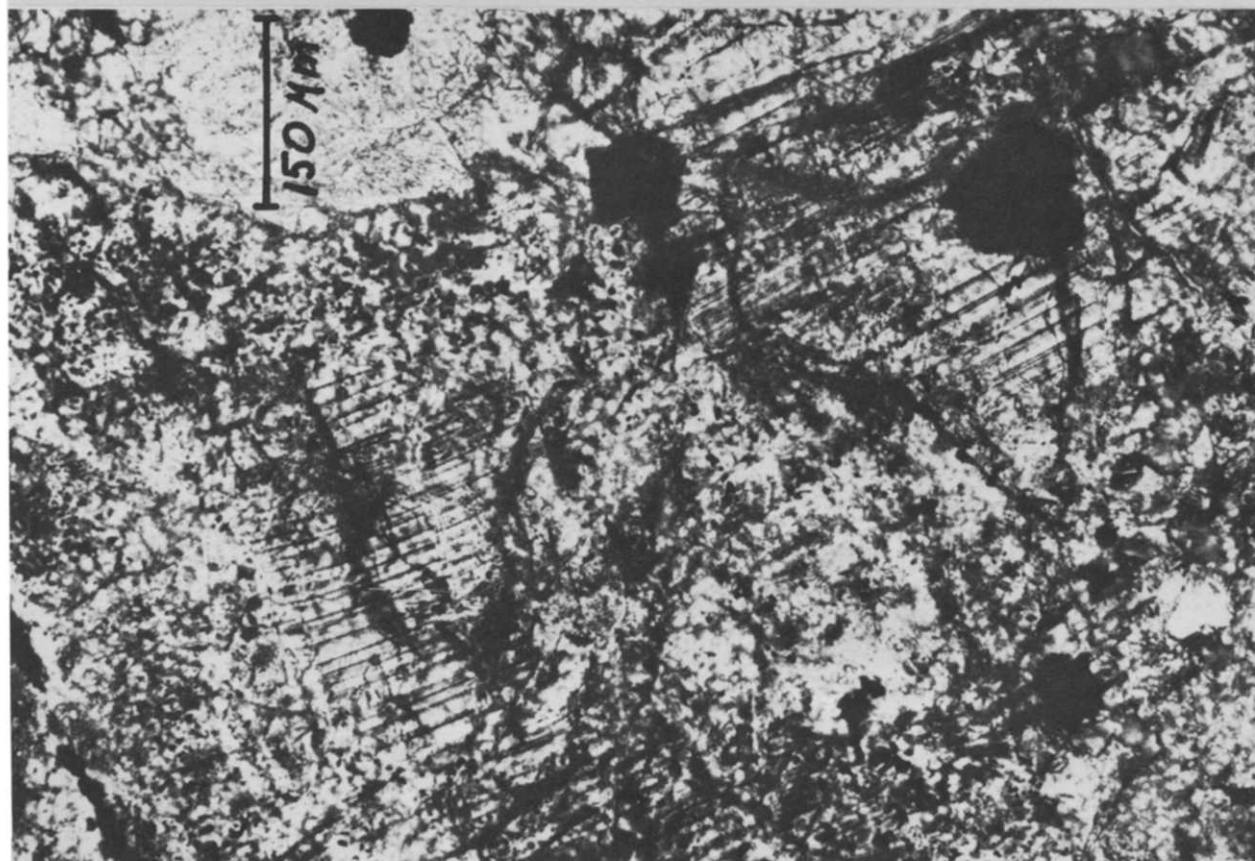


Fig. 2. Textural and mineralogical features of rock 12013 discussed in the text. Dimensions in parentheses refer to the diameter of the white circle. (a) K X-ray scan across dark-light lithology contact at 10/L showing K enrichment in light lithology (left half). The alkali feldspar penetrates into the dark lithology (center) (25 μm). (b) Secondary electron image of dark fragment at 12/K displaying micrographic intergrowth of plagioclase (dark), pyroxene (grey), and ilmenite (white) (10 μm). (c) Secondary electron image of lunar chondrule at 31/M-N. Plagioclase (dark), pyroxene (grey), ilmenite and titanite chromite (white) (80 μm). (d) Ca X-ray scan of low-Ca pyroxene with high-Ca pyroxene exsolution lamellae (7 μm). (e) K X-ray scan across fractured plagioclase at 24/R, showing K-rich vein filling (25 μm). (f) Ca X-ray scan across granitic vein at 43/N showing contrast between P-poor granitic material (center) and P-rich dark lithology (upper right) (50 μm). (g) P X-ray scan across dark-light lithology contact at 24/R showing concentration of P in the dark lithology (30 μm). (h) P X-ray scan across granitic vein at 42/F (15 μm). (i) Zr X-ray scan of zircon crystal in dark lithology at 42/F (15 μm).





(Fig. 3(c).

Fig. 3(a)–(c). Textural and mineralogical features of 12013 discussed in text. (a) S-shaped lamellae in pyroxene at 21-22/J-K. (b) Plagioclase grain showing linear train of bubble-like inclusions at 31/I-J. (c) Devitrified, plagioclase-rich glass spherule at 43-45/D-E. Bulk analysis is given in table 2 (analysis 1).

A few widely scattered dark grains (e.g., 12/K, 20/K, 20/H-I) display a micrographic (eutectic?) intergrowth of very fine-grained plagioclase, pyroxene and ilmenite. The texture, shown in fig. 2b, most likely developed during rapid cooling of liquid droplets or devitrification of glass fragments. Several fragments with chondrule-like textures have been identified. Two of these lunar chondrules may be seen at 30-31/M-N. Less obvious fragments have also been identified at 30/O-P and 24/D. The characteristic pattern of radiating prisms is illustrated in fig. 2c. Augite and Ca-poor pyroxene, plagioclase, ilmenite and titanian chromite are present. Again, these textures are due to devitrification or quench crystallization of liquid droplets.

4. Dark lithology

For convenience, all dark areas are discussed together in this section. We are not certain, however, that they all compose a single lithic unit. In particular, the main dark area of fig. 1a is somewhat distinct from the other dark areas in texture and some mineralogical details (cf. section on mineralogy). The dark color is due in large measure to a fine-grained matrix which is richer in opaque minerals (and perhaps very small fragments of devitrified glass?) than the light lithology. As previously pointed out, the dark lithology has been permeated by the granitic material somewhat less than the light lithology. Nevertheless, there is much evidence of alkali feldspar and silica in the dark lithology in addition to veins and well defined ovoid fillings. Many crystal and lithic fragments in the dark as well as the light lithology show fractures filled by the granitic component. Figs. 2e and 2f show a healed fracture in a plagioclase crystal. The dark lithology shows some signs of bulk plastic flow, which are best displayed by the outlines of the dark area in fig. 1a. This most probably took place when hot matrix fragments of the dark breccia were still glassy. Most of the mineral species identified in the light lithology may also be found in the dark lithology. Comparison of mineral compositions are given in a separate section below.

Anorthositic fragments have been recognized in

the dark lithology. One such fragment, 1 mm in diameter, may be seen at 32/E. It is composed of approximately 80% fine grained calcic plagioclase and 20% pyroxene (Ca-rich and Ca-poor). The pyroxene grains are scattered throughout the anhedral plagioclase matrix and additional subhedral plagioclase laths are enclosed in the pyroxene. Another plagioclase-rich fragment, approximately 0.3 mm in diameter, is located at 29/I. A small (0.3 mm) chondrule-like fragment (29/G) has been identified in the dark seam cutting across fig. 1b. It is similar texturally and mineralogically to the lunar chondrule fragments in the light lithology. An area of plagioclase composition (cf. table 2) is located at 43-45/D-E. The rounded outline of the area and the variolitic texture shown in fig. 3 suggest devitrified plagioclase spherule similar to those described in the Apollo 11 sample. A vein-like distribution of light material may be seen within the dark seam which cuts fig. 1b. It is composed largely of plagioclase which is significantly more sodic than the rest of the 12013 plagioclase. Some alkali-feldspar plus silica is associated with this plagioclase. Certain granitic fragments (e.g., 24/H-I and 32/H) appear to be completely incorporated within the dark seam of fig. 1b. These may be continuous in the third dimension with the granitic material of the light lithology.

If one looks at the texture of the dark lithology long enough, a pattern of arcuate trails of crystal and lithic fragments emerges. These resemble the features described by McKay et al. [5] and Richardson et al. [6] and identified by them as lunar accretionary lapilli formed in a hot, turbulent, impact-generated cloud. We are not fully convinced of the presence of these features in 12013, and we invite the reader to judge for himself. They seem best developed in the dark lithology of section 12013, 14 (fig. 1c, areas 42-43/O-P and 42-43/L-M). McKay et al. believe that the accretionary lapilli are the dominant structural feature of the Apollo 11 microbreccias, and their arguments appear convincing to us. However, it must be stated that these features are subtle and have not yet been identified by other groups who have examined the microbreccias. Formation of the dark lithology (and perhaps the entire rock?) as an ignimbrite laid down by a lunar base surge cloud is not incompatible with our results at this stage.

5. Granitic component

Potassic feldspar and silica occur throughout the entire rock in intimate intergrowth (fig. 4a). Their association in cross-cutting veins, ovoid fillings, and veinlets in fractured mineral grains (fig. 2e) can only be explained in terms of fluid injection. The fluid may have been a true granitic magma or a hot rhyolitic glass phase. The present textures make it difficult to differentiate unequivocally between these two possibilities. Two relatively large (1 mm) potassic feldspar grains have been found in close proximity to the otherwise much finer-grained granitic component. One of these has small inclusions of chromian ulvöspinel and encloses grains of relatively sodic ($\approx \text{An}_{52}$) plagioclase. Its composition ($\text{Or}_{65}\text{Ab}_{26}\text{An}_9$) falls within the range of the more abundant smaller potassic feldspar crystals. These larger alkali feldspar grains may represent phenocrysts of the granitic magma. Large crystals of tridymite (1 mm) have also been found in contact with the finer-grained granitic component, but their textural relationship to the fine-grained granitic component is not particularly suggestive of a phenocryst-groundmass association. Much of the alkali feldspar occurs in the form of euhedral, stubby prisms up to 50 μm long. These occur within a matrix of slender needles of silica and finer, often clouded, anhedral alkali feldspar (fig. 4a). Crystallization of potassium granite liquids or glass is well known to be a slow process. In the complete absence of fluidizing components, they will not completely crystallize in several years [8]. Very slow cooling and/or the erst-while presence of a volatile phase might explain the observed textures.

The ovoid fillings seen in the dark lithology of fig. 1a represent the occurrence of the granitic component least contaminated with obvious inclusions. The average of large beam analyses in eight separate fillings is presented in table 1. The slightly low total is due to cracks in the section. The molecular norm of the 12013 granitic component reaffirms its predominantly biminerale character. Liquid of this composition is most readily explained as the residual product of fractional crystallization or as an initial melt phase (not necessarily equilibrium melting, and obviously predominantly involving silica and potassic feldspar phases). The former mechanism is already

Table 1

Electron probe analyses of salic component in ovoid fillings of dark lithology in sample 12013. High silica glass analyses [7] from Apollo 11 rocks are included for comparison.

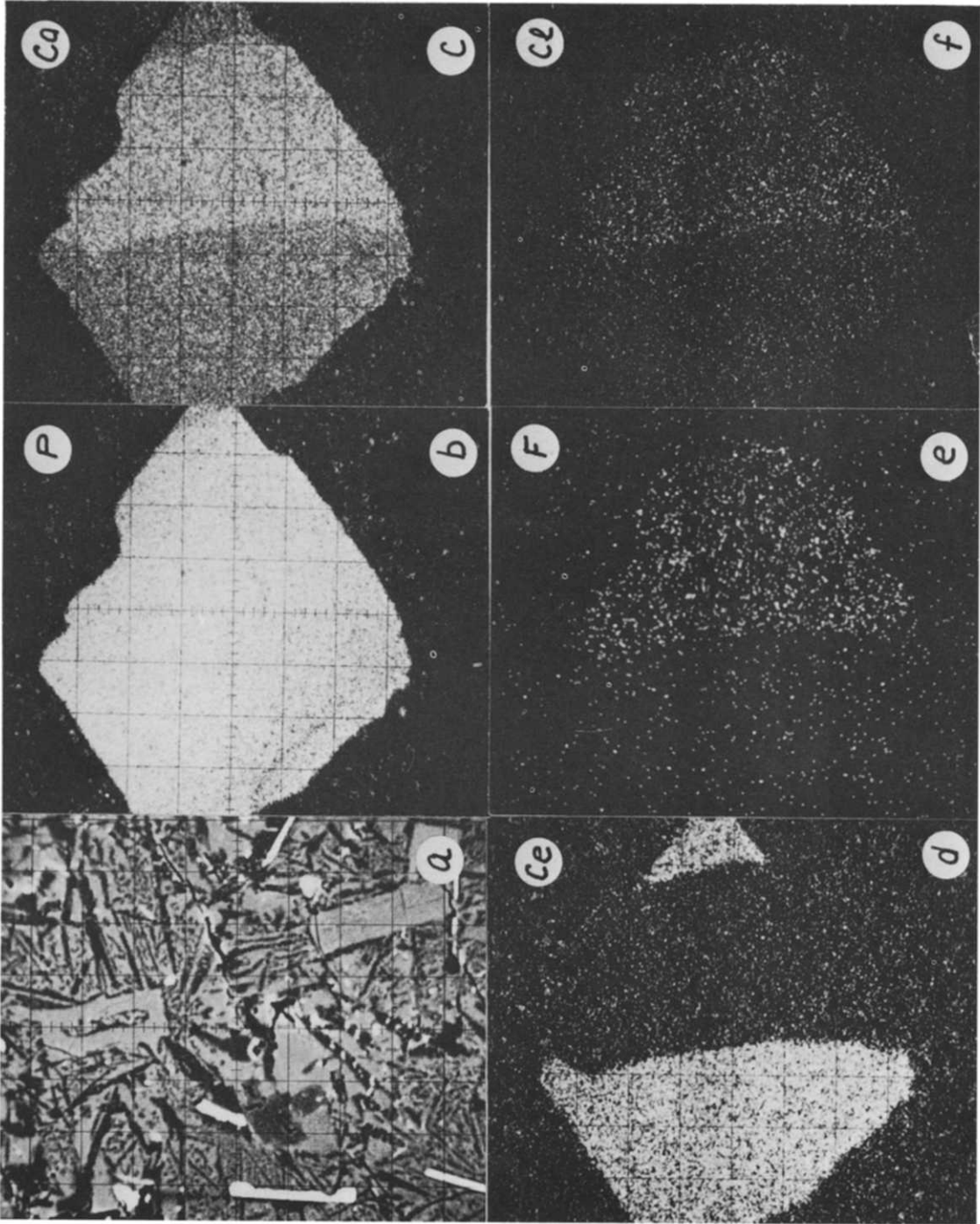
	Average of 8 fillings	s.d.	Average of 33 high Si glasses	s.d.
SiO_2	73.5	1.6	75.4	2.9
TiO_2	0.50	0.18	0.52	0.12
Al_2O_3	12.2	0.54	11.5	1.6
FeO	0.92	0.56	2.5	1.1
MgO	0.37	0.34	0.26	0.17
CaO	1.2	0.46	1.8	0.9
BaO	0.94	0.23	n.d.	n.d.
Na_2O	1.4	0.18	0.37	0.30
K_2O	6.9	0.58	6.5	2.4
	97.93		98.85	

Molecular norm of 12013 salic component:

Or = 43.0	97.1	Il = 0.7	2.9
Ab = 13.0		Wo = 0.3	
An = 5.3		En = 1.1	
Cn = 1.8		Fs = 0.8	
Q = 34.0			

known to result in the K-rich siliceous liquid now forming the mesostasis of many Apollo 11 igneous rocks. The average of 33 analyses of high-silica Apollo 11 mesostasis [7] is also given in table 1. The similarity in major element concentration is striking. There is significantly more Na and less Fe in the 12013 granitic phase. The difference in Fe is probably explained by unavoidable contamination of finely intermixed high-Fe glass in the Apollo 11 analyses [7]. Both compositions can quite accurately be termed potassic granites. Creation of a specialized, biminerale composition of this type by vapor phase differentiation seems less likely.

The presence of a granitic component in lunar rocks is not new, but the scale and concentration factors have been appreciably magnified in 12013. The probability of granitic rocks on the moon has been somewhat increased. It can be assumed that the dark microbreccia(s) and the fragmental components of the light lithology pre-date the injection of the granitic component. If the granitic component of 12013 is a portion of a larger granitic rock, several possible explanations exist for the final lithification



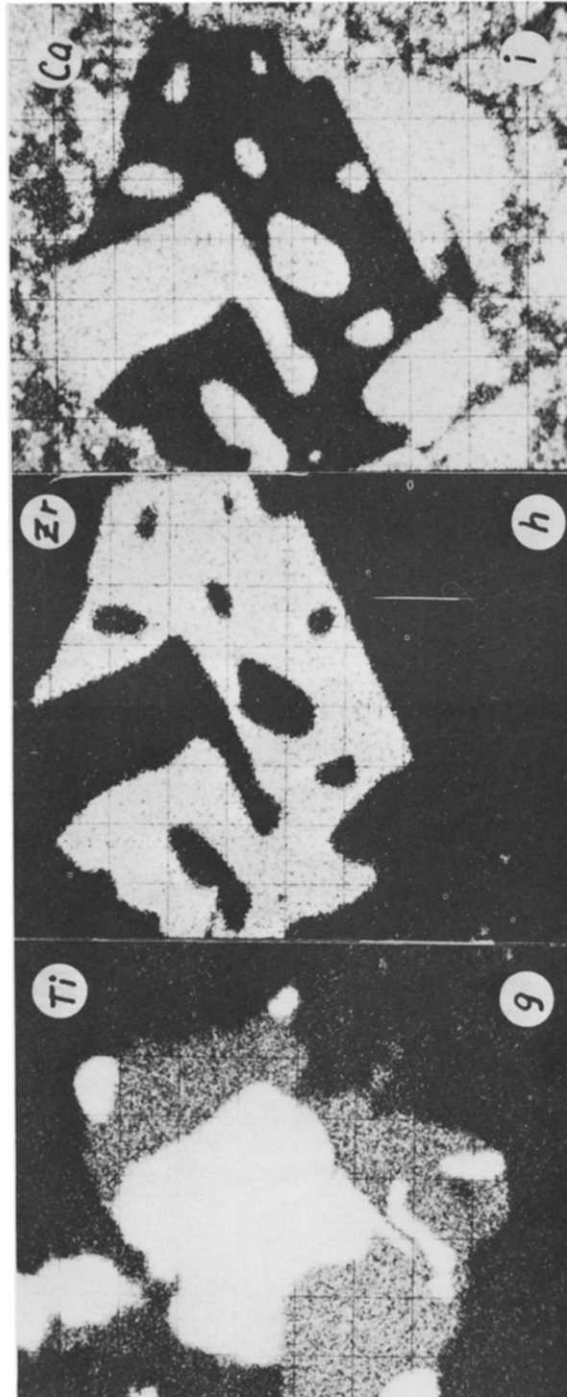


Fig. 4. Textural and mineralogical features of 12013 discussed in text. Dimensions refer to diameter of white circles. (a) BSE image of alkali feldspar (grey), silica (black) intergrowth in ovoid granitic area at 40-41/P-Q. Acicular ilmenite (white) is also present (20 μm). (b) through (f) X-ray scans of a composite phosphate (apatite-whitlockite) grain at 40/M, taken with P, Ca, Ce, F, and Cl radiation respectively (30 μm). (g) Ti X-ray scan of titanian chromite-ilmenite intergrowth in lithic fragment at 16-17/G-H (10 μm). (h) Zr X-ray scan of zircon-plagioclase intergrowth (15 μm). (i) Ca X-ray scan of same area as in (h).

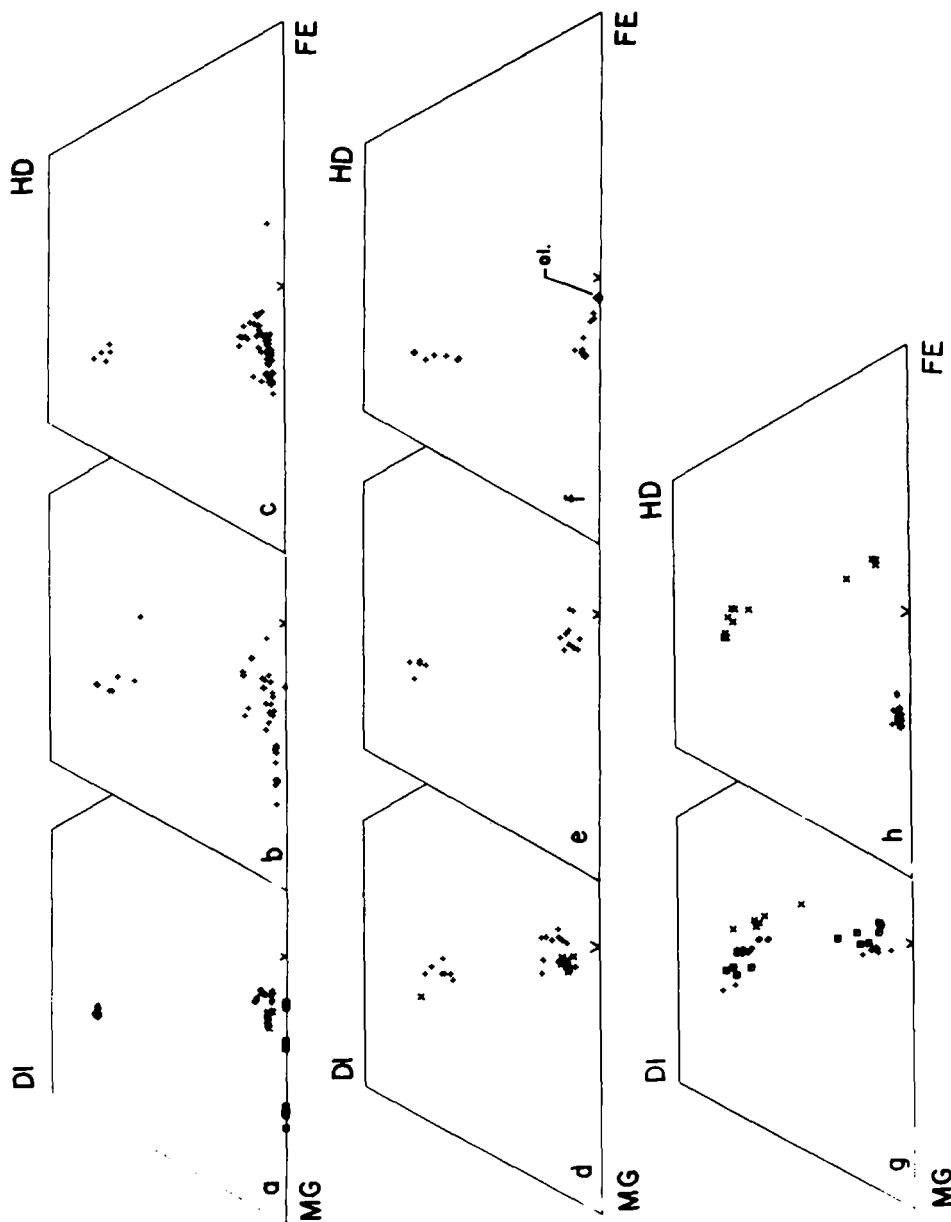


Fig. 5. Olivine and pyroxenes in 12013. (a) $\square$, three olivine crystal fragments from the dark lithologies of fig. 1a and c. Four pyroxene crystal fragments (different symbols) from the dark lithologies of fig. 1a and c. 46 points. (b) Pyroxene crystal fragments ($>15 \mu\text{m}$) in the dark lithology of fig. 1c. 42 points. (c) Pyroxenes ($<15 \mu\text{m}$) in the matrix of the dark lithology. 15 points. (d) Lithic fragments in the dark lithology. $\times$, 47-48/O-P; $+$, 10/N-P. 29 points. (e) Pyroxenes in the matrix of the light lithology. 15 points. (f) Olivine, Ca-poor pyroxene host and Ca-rich pyroxene lamellae in the ultramafic lithic fragment at 16-18/G-I. 31 points. (g) Pyroxenes associated with the granitic component. $+$ and $\square$, two areas in the light lithology; $\times$, pyroxene crystal at 34/E; $\square$, pyroxene crystal at 34/D. 34 points. (h) $+$, pyroxenes from area 41/O-P; $\times$, pyroxenes associated with alkali feldspar in the plagioclase-rich "vein" at 38-41/F-G. 41 points.

Table 2
Microprobe analyses of plagioclase, alkali feldspar, pyroxene, olivine, ilmenite, and titanian chromite in 12013.

	1	2	3	4	5	6	7	8	9	10	11	12
SiO ₂	48.82	51.30	60.93	51.55	35.95	0.21	0.27	0.26	0.37	0.28	0.27	0.31
TiO ₂	0.64	0.10	—	0.33	0.03	54.37	53.35	54.06	52.11	8.87	12.07	8.40
Al ₂ O ₃	30.32	31.35	22.52	0.59	0.17	0.17	0.23	0.20	0.10	10.17	6.26	8.50
Cr ₂ O ₃	—	—	—	0.23	0.03	0.19	0.61	0.28	0.50	41.61	40.25	43.61
FeO	1.14	0.21	0.13	26.15	35.99	41.29	43.27	43.48	44.56	36.54	37.64	36.92
MnO	—	—	—	0.35	0.33	0.33	0.35	0.39	0.35	0.28	0.35	0.34
MgO	0.65	0.10	0.0	19.09	28.23	3.49	2.58	2.18	1.39	2.60	2.23	2.10
CaO	14.59	13.66	3.87	1.73	0.12	0.10	0.08	0.19	0.14	0.08	0.12	0.21
BaO	n.d.	0.11	1.19	—	—	—	—	—	—	—	—	—
SrO	n.d.	n.d.	0.05	—	—	—	—	—	—	—	—	—
Na ₂ O	2.62	3.54	3.46	0.0	—	—	—	—	—	—	—	—
K ₂ O	0.34	0.34	7.99	—	—	—	—	—	—	—	—	—
Total	99.11	100.71	100.14	100.03	100.87	100.15	100.74	101.85	99.52	100.43	99.18	100.40

1. Plagioclase in devitrified glass spherule, 43-45/D-E.
2. Plagioclase crystal fragment in dark lithology.
3. Euhedral alkali feldspar in granitic component, 34/D.
4. Low-Ca pyroxene crystal fragment in dark lithology.
5. Olivine crystal fragment in dark lithology.
6. Ilmenite crystal fragment in dark lithology.

7. Ilmenite crystal fragment in light lithology.
8. Ilmenite in unaltered ophitic basalt lithic fragment, 23/O-P.
9. Acicular ilmenite in granitic material.
10. Titanian chromite crystal fragment in light lithology.
11. Titanian chromite from lunar chondrule in light lithology.
12. Titanian chromite in altered ophitic basalt lithic fragment.

stage of 12013. The non-granitic components may be xenoliths incorporated in a granitic magma. Alternatively, they may be fragments incorporated in a rhyolitic ignimbrite. In the latter case, the injection of the granitic component may have been in the form of rhyolite glass remobilized by compaction of the hot flow after it came to rest. A variant of the latter hypothesis might involve remobilization of an underlying rhyolitic regolith during compaction of a hot basic ignimbrite. Some of the vesicles in the dark lithology of fig. 1a are still voids and apparently the granitic fluid has not completely penetrated the microbreccia. This observation tends to favor the ignimbrite origin. Both hypotheses assume large volumes of granitic material. It is perhaps equally tempting to venture an explanation based on a more modest extrapolation not involving rock types that have not yet been identified in the current lunar sampling. A localized concentration of Apollo 11-type mesostasis is a possibility. There may be some mechanism, comparable to filter pressing, which can concentrate this low melting material from relatively small rock volumes and inject it into neighboring voids. Again, the compacting interior of a hot ignimbrite (in this case of more basic bulk composition) might be a suitable environment.

6. Mineralogy

The data presented in figs. 5 and 6 for the ferro-magnesian minerals and the feldspars represent three-element microprobe analyses. The raw data were corrected for background, drift, and matrix effects using correction factors obtained for each element from complete analyses of representative grains. These partial analyses are accurate to within 3%. Complete microprobe analyses of typical grains of the major phases are listed in table 2.

6.1. Olivine and pyroxene

Olivine and two pyroxenes occur in all lithologies. They range in size from submicron particles to 1 mm. For convenience, we have grouped the grains into crystal fragments ($> 15 \mu\text{m}$) and matrix ($< 15 \mu\text{m}$). The density of points in fig. 5 corresponds roughly to the relative abundance of the two pyroxenes. With the exception of the granitic component, low-Ca pyroxenes are predominant. Fig. 5 shows the rather restricted compositional range and limited iron enrichment of pyroxenes in 12013. We have not yet found pyroxenes showing the extreme iron enrichment of Apollo 11 pyroxenes. The few complete analyses

made indicate a significantly lower TiO_2 and Al_2O_3 content compared to the Apollo 11 pyroxenes. A notable feature of the pyroxene is the common occurrence of microprobe resolvable exsolution lamellae, which are found in both pyroxenes (fig. 2d). Analysed host-lamellae pairs from a large low-Ca pyroxene crystal in the ultramafic lithic fragment (16-18/G-I) are shown in fig. 5f.

Crystal fragments of olivine in the dark lithology are quite homogeneous, but a range of compositions is covered by the different grains (fig. 5a). Individual pyroxene crystal fragments in the dark lithology also show a restricted compositional range (fig. 5a) though the total spread is fairly large (fig. 5b). The more Mg-rich olivines and pyroxenes occur in the dark lithology of 12013, 14. Pyroxenes in the matrix of the dark lithology show a narrower compositional range (fig. 5c). This may be due to inadequate sampling. Pyroxenes in the matrix of the light lithology (fig. 5e) are significantly more iron-rich than those in the dark lithology, indicating a different source region.

A recurring problem in studying the granitic material is differentiating authigenic minerals from those incorporated from the surrounding material. Apart from alkali feldspar and silica needles, the same minerals occur in both. Criteria which we have found useful in making the distinction are crystal habit, chemical composition and the abundance of the particular mineral in the adjacent non-granitic material, although it is possible that the intrusive granitic component has incorporated xenocrysts from another locality.

Small rounded grains of pyroxene (fig. 5g, symbols + and □) in the granitic material may be contaminants, since they are not significantly different in composition from pyroxenes in the adjacent material. Two elongated prisms of Ca-rich pyroxene are associated with the granitic material at 34/E-D. This crystal habit is not found in pyroxenes elsewhere in 12013, leading us to believe they are authigenic. This conclusion is reinforced by their relatively iron-rich composition (fig. 5g). Similar pyroxenes are associated with the granitic material at 38-41/F-G (fig. 5h).

6.2. Feldspars

Selected data from several hundred feldspar analyses are presented in fig. 6. The total range of plagioclase composition is An_{45} to An_{97} . This wide

range is observed in the matrix ($< 15 \mu\text{m}$) of both light and dark lithologies. Plagioclase crystal fragments larger than $15 \mu\text{m}$ define a similar range of composition in the light lithology (fig. 6a) but do not appear to extend as far to the sodic end in the dark lithology (fig. 6c). This probably reflects the small sampling of only four individual grains. Fig. 6c indicates that these individual crystals are relatively homogeneous and unzoned. Certain groupings of plagioclase and pyroxene crystals form arcuate trails which may be related to accretionary lapilli formation. The compositions of plagioclase and pyroxene from these areas show little variation, suggesting a common provenance. Analyses from such an area at 41/O-P are shown in figs. 5h and 6i. Apollo 11 igneous rocks contain very few plagioclases more sodic than An_{70} . The igneous rock plagioclase range for Apollo 12 is An_{50} to An_{90} [1], more nearly compatible with that found in 12013.

A light colored, irregularly shaped vein occurs within the dark seam cutting across fig. 1b. It is composed dominantly of plagioclase, but contains local concentrations of alkali feldspar and silica. The plagioclase composition is constant and significantly more sodic (An_{50} to An_{55}) than elsewhere in 12013. Fig. 6g shows a composition plot of this plagioclase together with the associated alkali feldspar which suggests a possible equilibration between the two solid solutions.

Plagioclase compositions from various lithic fragments and lunar chondrules (figs. 6e, 6f and 6h) indicate little compositional variation within individual fragments.

Alkali feldspar is the major phase of the granitic component which has permeated much of rock 12013. The total range of composition is given in fig. 6j along with the approximate limit of solid solution expected for high temperature, dry magmas based upon terrestrial feldspars [8]. A large number of the lunar alkali feldspars significantly exceed this solubility limit. Many of these analyses refer to relatively large crystals which appear quite homogeneous within the limit of resolution of the microprobe and optical microscope (fig. 4a). A complete analysis of an An-rich alkali feldspar is given in table 2. BaO is present in the alkali feldspars in concentrations of up to 4 wt.%, with the highest concentrations being found in the most potassic varieties.

Analyses at $2 \mu\text{m}$ steps across a single alkali feldspar

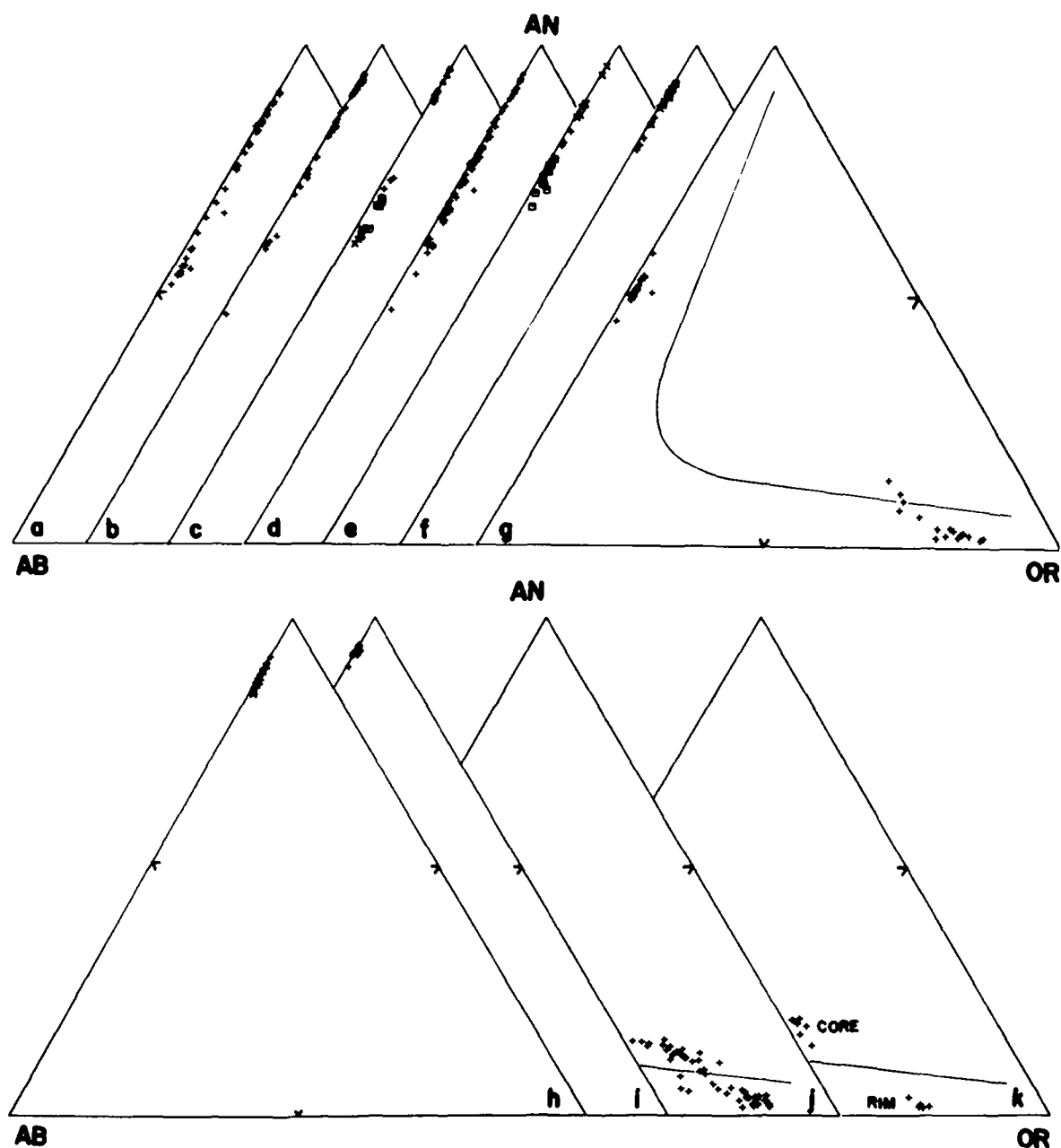


Fig. 6. Plagioclase and alkali feldspar in 12013. (a) Random plagioclase crystal fragments ($> 15 \mu\text{m}$) in the light lithology. 37 points. (b) Random plagioclase crystal fragments ($< 15 \mu\text{m}$) in the matrix of the light lithology. 36 points. (c) Five plagioclase crystal fragments ($> 15 \mu\text{m}$) (different symbols) in the dark lithology of fig. 1a. 25 points. (d) Random plagioclase crystal fragments ($< 15 \mu\text{m}$) in the matrix of the dark lithology. 58 points. (e) Plagioclase from three lithic fragments in the dark lithology. x and +, anorthositic fragments at 31-32/E-F and 29/H-I respectively; $\square$, 10/N-P. 22 points. (f) Plagioclase from three chondrules in the light lithology. x, 31/M-N; +, 30/N; $\square$, 29-30/L-M. 16 points. (g) Plagioclase and alkali feldspar from plagioclase-rich area, 38-41/F-G. 38 points. (h) Plagioclase from two ophitic basalt lithic fragments in the light lithology. x, 23/O-P; +, 18-19/M-N. 13 points. (i) Plagioclases from area 41/O-P. 19 points. (j) Alkali feldspars from many areas of granitic material (all lithologies). 50 points. (k) Analyses (2 μm steps) across a euhedral crystal of alkali feldspar (cf. fig. 4a). 12 points.

crystal are shown in fig. 6k. A sharp drop in An content occurs a few μm from the edge. In general, euhedral crystals are richer in An content than fine-grained matrix alkali feldspars, suggesting that An-rich material crystallized prior to the An-poor material.

The total analyses of both plagioclase and alkali feldspars completed to date indicate a close approach to stoichiometric requirements. This contrasts with analyses of Apollo 11 plagioclases from certain igneous rocks [10, 11].

6.3. Ilmenite

Ilmenite, the most abundant opaque mineral, is found in varying proportions in all lithologies. Abundant, fine-grained ilmenite in the matrix of the dark lithology is largely responsible for the dark color of these areas. There is a complete gradation in grain size from millimeter-size to submicron-size crystal fragments. A typical analysis of ilmenite from the dark lithology matrix is given in table 2.

Ilmenite is much less abundant in the light lithology where it occurs as rather large crystal fragments in the matrix (analysis 7). All clearly identified lithic fragments in the light lithology contain variable amounts of ilmenite. The range of composition is limited, and only one typical analysis (no. 8) is shown. Lunar chondrules (fig. 2c) and the fine-grained micrographic intergrowths (fig. 2b) contain small ($< 10 \mu\text{m}$) equant ilmenite grains with a virtually constant composition.

The plagioclase-rich vein which cuts the dark lithology in the upper part of fig. 1b, is almost free of ilmenite. The few grains found may represent non-authigenic ilmenite from the surrounding dark lithology. Equant grains of ilmenite (composition similar to no. 7) make up approximately one per cent of the anorthositic lithic fragments in the dark lithology. The granitic material contains minor amounts of Mg-poor, authigenic ilmenite with an acicular habit (fig. 4a, analysis no. 9).

All ilmenites are stoichiometric with $(\text{Fe} + \text{Mg} + \text{Mn}) = \text{Ti} = 1.00 \pm 0.02$ (3 oxygens). SiO_2 , CaO , Cr_2O_3 , Al_2O_3 comprise approximately one weight per cent of the total analysis. The most important compositional variable is the $\text{Mg}/(\text{Mg} + \text{Fe})$ mole ratio. The average value of this ratio is 0.138 (dark lithology), 0.098 (light lithology matrix), 0.082 (ophitic basalt),

0.086 (chondrules), and 0.052 (granitic material). These data are in harmony with the pyroxene data which show a similar trend in the $\text{Mg}/(\text{Mg} + \text{Fe})$ ratio. The composition of ilmenite in the granitic material is consistent with the highly differentiated character of the latter.

6.4. Titanian chromite, aluminian chromite, and chromian ulvöspinel

Although Cr in 12013 is lower than in other Apollo 12 and Apollo 11 samples [1], chromium-rich spinels are relatively abundant. There is a considerable range in composition but few are as titanium-rich as those reported from Apollo 11 crystalline rocks and breccias [9]. Representative analyses from various lithologies are given in table 2. Chromites are relatively rare in the matrix and lithic fragments of the dark lithology. While no large crystal fragments of chromite were found, there are some micron-size grains in the dark matrix. Large ($200 \mu\text{m}$ to $300 \mu\text{m}$) fragments of chromite are scattered throughout the light lithology (no. 10). The altered ophitic basalt fragments in the light lithology contain chromites with a range in TiO_2 of 7.2 to 9.2 wt.%. No chromite was found in the unaltered ophitic basalt. Lunar chondrules contain equal amounts of chromite ($\text{TiO}_2 = 5.3$ to 14.0) and ilmenite (fig. 2c), while in some of the coarse-grained lithic fragments, e.g. 21-22/J, chromite ($\text{TiO}_2 = 3.7$ to 12.1) is the predominant opaque mineral.

Chromites are absent from the plagioclase-rich vein which cuts the dark lithology in the upper part of fig. 1b. With the exception of the large crystal of alkali feldspar at 33/F, which contains a few inclusions of chromian ulvöspinel ($\text{TiO}_2 = 25.0$ to 26.2), no spinels were found in the salic material. Composite crystal fragments of ilmenite and titanian chromite are common in some coarse-grained lithic fragments (fig. 4g).

Apart from the ulvöspinel, the spinels are relatively rich in the chromite molecule. The range of composition is from $\text{Chr}_{51.5}\text{Usp}_{38.9}\text{Sp}_{9.6}$ to $\text{Chr}_{70.5}\text{Usp}_{14.4}\text{Sp}_{15.1}$ where $\text{Chr} = \text{FeCr}_2\text{O}_4$, $\text{Usp} = \text{Fe}_2\text{TiO}_4$, and $\text{Sp} = (\text{Fe}, \text{Mg}, \text{Mn})\text{Al}_2\text{O}_4$. $\text{SiO}_2 + \text{CaO}$ is less than 0.5 wt.%.

6.5. Troilite

Troilite is not abundant. A systematic search reveals a few micron-size grains of troilite distributed throughout the dark lithology, the light lithology, and lithic fragments within these lithologies. An ex-

ception to this general rule is one lithic fragment at 47-48/O-P in the dark lithology composed almost entirely of low-Ca pyroxene with several per cent troilite and associate minor iron metal.

In contrast to the paucity of troilite in the dark and light lithologies, it is relatively abundant in the salic material. Crystals up to 50 μm make up a few per cent of the granitic vein material at 22-23/Q-S. Several of these larger troilite grains were analysed for Fe and S. A typical analysis gives Fe = 63.3 and S = 35.5 wt.%. This corresponds to a mole ratio Fe/S = 1.02, and it is possible that approximately 2% Fe metal is included in the troilite as submicroscopic grains.

6.6. Iron metal

The largest concentrations of iron are associated with troilite, largely in the salic material and the single lithic fragment described above. Although composite troilite-iron grains are found, the sponge texture of troilite-iron intergrowths common in Apollo 11 samples was not observed. Iron is greatly subordinate to troilite and forms an extremely small fraction of the total rock. It is of interest, however, because of its very high Ni content. Three grains analysed gave a range of 26.7 to 27.6 wt.% Ni. These metal grains in the granitic material are not derived from the rest of 12013, and were introduced into the rock by the granitic component, i.e., they are not of direct meteoritic origin. The possibility of recycling meteoritic iron indicates that caution must be applied when using high-Ni iron grains in calculating meteoritic contributions to the lunar regolith.

6.7. Apatite and whitlockite

Systematic examination reveals that phosphates are present in all areas of 12013. They are most abundant in the dark lithology where crystal fragments from submicron-size to 300 μm are found. A P X-ray scan across the dark-light lithology contact (fig. 2g) clearly shows the relative abundance of phosphates in these areas. Outside the dark lithology, localized concentrations are found, e.g., in the plagioclase-rich material at 33/G.

The typical occurrence of the phosphate minerals is as composite grains, with an intergrown high-Ca and a lower-Ca phosphate. Partial analyses (uncorrected) of these two phosphates are shown in table 3. While final determination as to mineral type must

await detailed analysis and X-ray data, it appears that the high-Ca phosphate is a chloro-fluorapatite with a low concentration of REE, Fe, Si, and Mg. The low-Ca phosphate is similar to the "whitlockite" described by Albee and Chodos [3]. A qualitative comparison between the two parts in a large composite crystal fragment (40/M) is shown in figs. 4b-4f. The most striking contrasts are shown by Ce and F. Fluorine is very low in the whitlockite but a trace of chlorine is present. The whitlockite contains approximately 10 per cent $\text{RE}_2\text{O}_3 + \text{Y}_2\text{O}_3$ (calculated by difference).

Table 3
Partial (uncorrected) analyses of apatite and whitlockite

	1	2
SiO_2	0.3	0.4
FeO	0.2	1.2
MgO	0.2	2.8
CaO	55.2	40.6
P_2O_5	41.9	41.8
F	3.7	0.1
Total	101.5	86.7
O = F	1.6	
Total	99.9	

1. Chloro-fluorapatite from a composite phosphate crystal fragment at 40/M. The CaO, P_2O_5 , and F values were obtained by comparison with an apatite standard and are essentially correct.
2. Whitlockite from the same grain as 1.

Phosphates are not abundant in the salic material as can be seen in fig. 2h which shows a P X-ray scan across the contact between dark lithology and a salic vein at 43/N. However, minor amounts of both phosphates do occur in the salic material, and acicular crystals of whitlockite are almost certainly authigenic. Complete analyses are not available at this time.

The intergrowth of the apatite and whitlockite is unusual. The contact is sharp and fairly straight (fig. 4d). None of the grains is large enough to determine whether one phosphate forms a core and the other a rim but many grains, e.g., the grain in fig. 4b-f, show whitlockite partially enclosing apatite which may be interpreted as whitlockite rimming an apatite core.

6.8. Zircon

Sample 12013 contains 2200 ppm Zr [1]. If all the Zr is concentrated in zircon, there should be approximately 0.1 % zircon in the mode. Most zircon grains occur in the dark lithology in grains ranging from $< 1 \mu\text{m}$ to $80 \mu\text{m}$. Fig. 2i is a Zr X-ray scan of a large zircon fragment in the dark lithology. An unusual grain showing zircon-plagioclase intergrowth in the dark lithology is shown in figs. 4h and 4i.

Zircon can also be found in the light lithology, usually in grains $< 10 \mu\text{m}$. Small zircon crystals are relatively common in the salic material. Their occurrence in association with the alkali feldspar-silica matrix, isolated from obviously non-authigenic fragments, indicates that they may have crystallized at the same time as the granitic material. Complete quantitative analyses are not yet available, but comparisons with a zircon standard indicate that it is virtually pure ZrSiO_4 . The Hf/Zr ratio has not yet been determined but since the Hf/Zr in the bulk rock is close to 0.01, Hf is probably present in only minor amounts in the zircon. X-ray area scans reveal low concentrations of REE.

6.9. Silica minerals

In addition to the acicular silica (polymorph unknown) associated with the alkali feldspar in the granitic component, other silica minerals with a different habit have been found. Large crystal fragments of tridymite (large positive 2 V) up to 1 mm occur in the dark lithology, e.g., at 39/O-P and 40/K. These grains contain areas of variable extinction which give rise to a mosaic texture. They are always surrounded by a narrow rim of the granitic material.

Small, rounded crystals of quartz (uniaxial positive, $\epsilon - \omega > 0.006$) up to $200 \mu\text{m}$ are found in some areas of the granitic material, e.g., at 26/S. They are enclosed by a rim of additional silica material (identified by contrast in cathodoluminescence). Projecting from their rim into an alkali feldspar matrix are the more common acicular silica crystals. The quartz cores contain more than 99% SiO_2 , while the maximum observed SiO_2 content of the acicular crystals is 95%. This low value may be due to the difficulty in resolving the fine needles with the probe beam. The textural relations suggest that the larger quartz grains were initially resorbed by a silica undersaturated granitic liquid and subsequently acted as sites for silica crystallization.

7. Final comments (modesty clause)

Our analytical results are not final, and we anticipate that additional work will not only refine the mineralogical data, but also lead to alternate genetic interpretations. It is quite intentional that this section has not been entitled "conclusions". We have attempted to point out some of the possibilities for explaining certain features of this rock in previous sections, but to draw any firm or general conclusions about as complex a piece of matter as rock 12013 would be pretentious at this stage. The natural desire of petrologists to explain rock formation in terms of one or two well defined events in time and space must probably be abandoned when dealing with this mixed bag. The preliminary survey indicates a greater diversity of fragmental components and of rock forming processes than heretofore encountered in lunar samples. This must be kept in mind while interpreting the geochemical data.

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